Cell Biology

Structure, Function & Division — Introductory Module

Sample coursework material — for product demonstration purposes.

Chapter 1 — What Is a Cell?

Every living organism, from a single bacterium to a blue whale, is built from cells. The cell is the smallest unit that can be considered alive: it can take in nutrients, convert them into energy, carry out specialized functions, and reproduce.

Cells fall into two broad categories. Prokaryotic cells, found in bacteria and archaea, have no nucleus and a relatively simple internal structure. Eukaryotic cells, found in plants, animals, fungi, and protists, are larger and organize their contents into membrane-bound compartments called organelles. This module focuses on eukaryotic, and specifically animal, cell structure.

- All cells arise from pre-existing cells (cell theory).
- The cell is the basic structural and functional unit of life.
- Eukaryotic cells compartmentalize work into organelles; prokaryotic cells do not.

Chapter 2 — The Cell Membrane and Cytoplasm

The cell membrane is a thin, flexible boundary made of a phospholipid bilayer studded with proteins. It is selectively permeable, meaning it controls what enters and leaves the cell — allowing oxygen and nutrients in while keeping waste products contained until they can be exported.

Inside the membrane lies the cytoplasm, a gel-like fluid that suspends the organelles and provides the medium in which most chemical reactions of the cell take place. The cytoplasm also contains the cytoskeleton, a scaffold of protein fibers that gives the cell its shape and allows organelles to move within it.

Chapter 3 — The Nucleus and Genetic Control

The nucleus is the cell's control center. It houses the cell's DNA, organized into chromosomes, and is enclosed by its own double membrane called the nuclear envelope. Pores in this envelope allow molecules like messenger RNA to pass between the nucleus and the cytoplasm.

Inside the nucleus, a dense region called the nucleolus assembles ribosomal subunits, which are later shipped out to the cytoplasm to build ribosomes — the cell's protein-manufacturing machinery. Because the nucleus controls which genes are switched on or off, it ultimately determines what proteins a cell produces and what job that cell performs in the body.

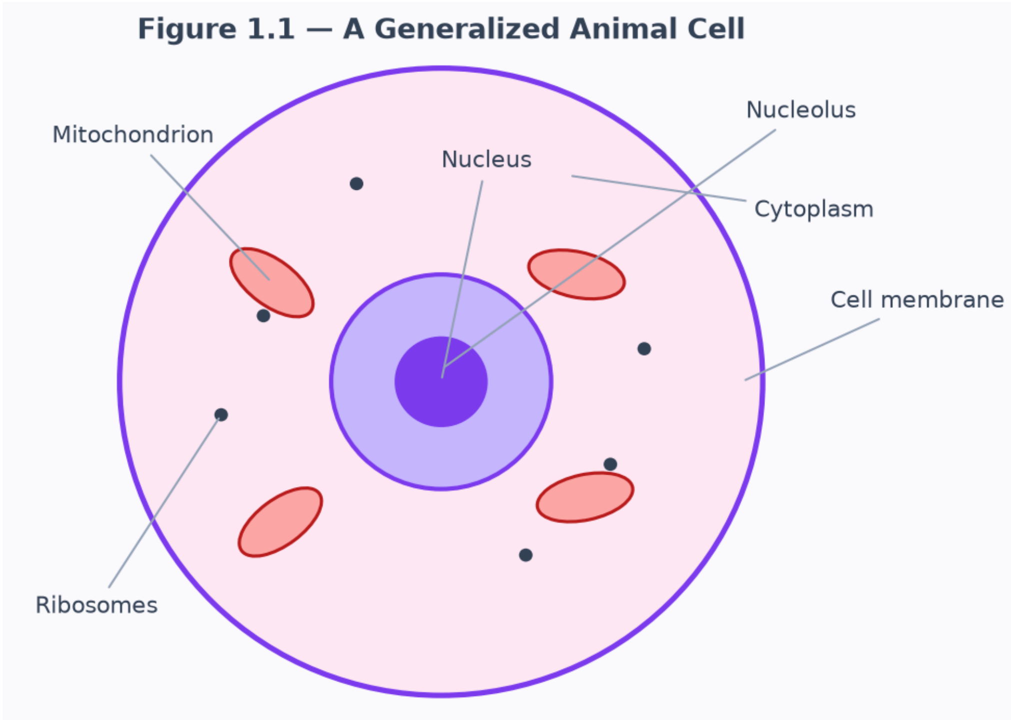

Figure 1.1 — A generalized animal cell, showing the nucleus, nucleolus, mitochondria, ribosomes, and cell membrane.

Chapter 4 — Mitochondria and Energy Production

Mitochondria are often called the powerhouses of the cell. Through a process called cellular respiration, they convert the chemical energy in glucose into adenosine triphosphate, or ATP — the molecule cells use to fuel nearly every energy-requiring process, from muscle contraction to active transport across membranes.

Mitochondria have their own small loop of DNA and can replicate independently of the rest of the cell, a fact that supports the endosymbiotic theory: the idea that mitochondria were once free-living bacteria that were engulfed by an ancestral cell billions of years ago and formed a permanent, mutually beneficial partnership.

Chapter 5 — The Cell Cycle

Cells do not divide continuously. Instead, they progress through an ordered sequence of stages called the cell cycle. Most of a cell's life is spent in interphase, which is subdivided into a growth phase (G1), a DNA synthesis phase (S) in which the chromosomes are duplicated, and a second growth phase (G2) in which the cell prepares the proteins it will need to divide.

Only after interphase is complete does the cell enter the mitotic (M) phase, where it physically divides its duplicated chromosomes and cytoplasm into two identical daughter cells. Checkpoints between each stage verify that DNA has been copied correctly before the cell is allowed to proceed — a safeguard that, when it fails, can lead to uncontrolled division and cancer.

Chapter 6 — Mitosis: Step by Step

Mitosis is the stage of the cell cycle in which a cell's duplicated chromosomes are separated into two identical nuclei. It is conventionally divided into four phases.

During prophase, the duplicated chromosomes condense into visible X-shaped structures and the mitotic spindle begins to form. In metaphase, the chromosomes line up along the cell's equator, attached to spindle fibers from opposite poles. Anaphase follows, as the spindle fibers shorten and pull the sister chromatids apart toward opposite ends of the cell. Finally, in telophase, two new nuclear envelopes form around each set of chromosomes, and the cell pinches in two during cytokinesis — producing two genetically identical daughter cells.

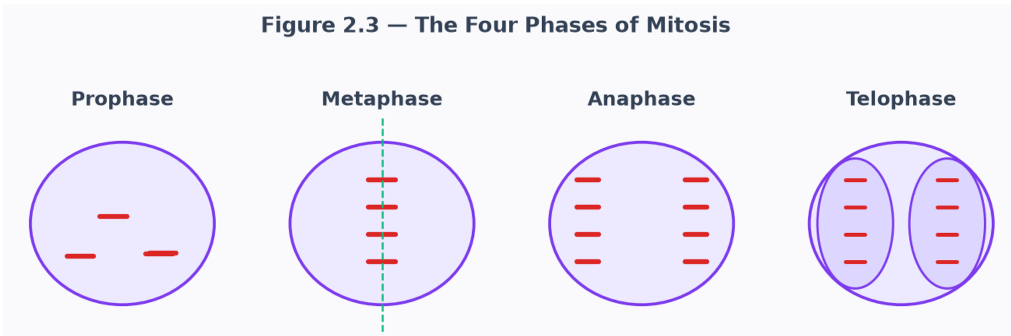

Figure 2.3 — The four phases of mitosis: prophase, metaphase, anaphase, and telophase.